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ENBC 332

Homework 4 Discussion and Outputs

Problem 1 Part C:

The sign of covariance indicates the direction of the linear relationship between the two variables of age and charges. The covariance I calculated for the dataset that I originally used in homework 2 was 50874.8 which indicates that they move in the same direction linearly. However, if the covariance was found to be negative, it would indicate that both variables are moving in the opposite direction of each other.

Pearson's correlation coefficient is a method of measuring the linear correlation between two sets of variables. More specifically, the formula used derives a number between negative one to positive one where a more negative value returned suggests that there is a negative linear relationship between both variables while a positive value would represent vice versa. The magnitude of the coefficient is also an indicator of how strong that relationship between both variables is. More specifically, the correlation is perfect if the coefficient is between 1 and 0.9, strong between 0.9 and 0.5, weak between 0.5 and 0.1, and uncorrelated below that.

For the dataset used from homework 2, I found that Pearson's correlation coefficient was approximately 0.2990082 which suggests a weak positive linear relationship between both variables.

As shown, the difference between Pearson's correlation coefficient and covariance is that the coefficient can only show how strong the linear relationship between two variables is positive or negative. Covariance is different where it can only provide information on whether the two trends given by both variables move in the same direction or in the opposite direction of each other.

Outputs:

Problem 1:

```

The downloaded binary packages are in
/var/folders/tn/tj4_th0577vb43zq_58mwbm0000gn/T//RtmpEoLuP7/downloaded_packages
> library(readxl)
> data <- read_excel('/Users/chrislee/Desktop/R Workspace/ENBC 332 - Homework2/ENBC 332 - HW02 Problem1 Excel File.xlsx', sheet = "Sheet1")

> # Part A
> covar_for <- function(temp_data) {
+   total_dev <- 0
+   mean_age <- mean(temp_data$age)
+   mean_charges <- mean(temp_data$charges)
+   col_length <- length(temp_data$age)
+   for (i in 1:col_length) {
+     total_dev <- total_dev + (temp_data$age[i] - mean_age) * (temp_data$charges[i] - mean_charges)
+   }
+   print(total_dev)
+   covariance <- total_dev / (col_length - 1)
+   return(covariance)
+ }
> covar_sum <- function(temp_data) {
+   total_dev <- 0
+   mean_age <- mean(temp_data$age)
+   mean_charges <- mean(temp_data$charges)
+   col_length <- length(temp_data$age)
+   total_dev <- sum((temp_data$age - mean_age) * (temp_data$charges - mean_charges))
+   print(total_dev)
+   covariance <- total_dev / (col_length - 1)
+   return(covariance)
+ }
> cat("Covariance for dataset using for loop:", covar_for(data))
[1] 68019611
Covariance for dataset using for loop: 50874.8
> cat("Covariance for dataset using sum function:", covar_sum(data))
[1] 68019611
Covariance for dataset using sum function: 50874.8

> # Part B
> ### Previous functions from homework 2
> var_func_for <- function(data_col) {
+   data_mean <- mean(data_col)
+   total_subtracted_mean <- 0
+   for (data_point in data_col) {
+     total_subtracted_mean <- total_subtracted_mean + (data_point - data_mean)^2
+   }
+   data_variance <- total_subtracted_mean / (length(data_col) - 1)
+   return(data_variance)
+ }
> var_func_sum <- function(data_col) {
+   data_mean <- mean(data_col)
+   subtracted_mean_values <- (data_col - data_mean)
+   squared_difference <- subtracted_mean_values^2
+   total_subtracted_mean <- sum(squared_difference)
+   data_variance <- total_subtracted_mean / (length(data_col) - 1)
+   return(data_variance)
+ }
> sd_func_for <- function(data_col) {
+   data_mean <- mean(data_col)
+   total_subtracted_mean <- 0
+   for (data_point in data_col) {
+     total_subtracted_mean <- total_subtracted_mean + (data_point - data_mean)^2
+   }
+   data_standard_deviation <- sqrt(total_subtracted_mean / (length(data_col) - 1))
+   return(data_standard_deviation)
+ }
> sd_func_sum <- function(data_col) {
+   data_mean <- mean(data_col)
+   subtracted_mean_values <- (data_col - data_mean)
+   squared_difference <- subtracted_mean_values^2
+   total_subtracted_mean <- sum(squared_difference)
+   data_standard_deviation <- sqrt(total_subtracted_mean / (length(data_col) - 1))
+   return(data_standard_deviation)
+ }
> cor_coe_for <- function(temp_data) {
+   cor_coe <- covar_for(temp_data) / (sd_func_for(temp_data$age) * sd_func_for(temp_data$charges))
+   return(cor_coe)
+ }
> cor_coe_sum <- function(temp_data) {
+   cor_coe <- covar_sum(temp_data) / (sd_func_sum(temp_data$age) * sd_func_sum(temp_data$charges))
+   return(cor_coe)
+ }
> cat("Pearson's correlation coefficient for the dataset using for loops:", cor_coe_for(data))
[1] 68019611
Pearson's correlation coefficient for the dataset using for loops: 0.2990082
> cat("Pearson's correlation coefficient for the dataset using sum function:", cor_coe_sum(data))
[1] 68019611
Pearson's correlation coefficient for the dataset using sum function: 0.2990082

> # Part C
> plot(data$age, data$charges,
+       main="Scatter Plot of Coffee Charges in Relation to Age",
+       xlab="Age",
+       ylab="Charges",
+       pch=19)
> |

```

Scatter Plot of Coffee Charges in Relation to Age

